

Chapter 1, Problem 6PP

Problem

A home electric heater draws 10 A when connected to a 115 V outlet. How much energy is consumed by the heater over a period of 6 hours?

Step-by-step solution

Step 1 of 2

A home electric heater, when connected to a 115 V outlet, draws 10 A .

Calculate the power of the electric heater.

$$\begin{aligned} p &= vi \\ &= (115)(10) \\ &= 1150 \text{ W} \end{aligned}$$

Step 2 of 2

The energy consumed by the heater over a period of 6 hours is,

$$\begin{aligned} w &= pt \\ &= (1150)(6) \\ &= 6.9 \text{ kWh} \end{aligned}$$

Therefore, the energy consumed by the heater over a period of 6 hours is **6.9 kWh** .